

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

ERROR ANALYSIS TASK

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COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 30%

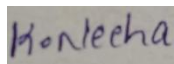
Duration: 30 minutes

Marks: 25

Code of Conduct:

I K.Neeha(192425245)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS:

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

a) Identify the conceptual error in the likelihood formulation.

Answer :

The conceptual error is that the likelihood function is written as the sum of probabilities instead of the product of probabilities of all independent observations. In MLE, the likelihood

represents the joint probability of the observed data given the model parameters, so probabilities must be multiplied, not added.

b) Explain how this mistake affects parameter estimation.

Answer :

Using the sum of probabilities does not correctly represent the probability of the complete dataset. As a result:

- The optimization objective becomes incorrect.
- The estimated parameters do not maximize the true likelihood.
- The model may produce biased or inaccurate parameter values.
- Prediction accuracy decreases even with a large amount of training data.

c) Suggest the correct mathematical formulation and reasoning.

Answer :

For independent observations $x_1, x_2, \dots, x_n$, the correct likelihood function is:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

The corresponding log-likelihood is:

$$\log L(\theta) = \sum_{i=1}^n \log P(x_i | \theta)$$

Reasoning:

The product represents the joint probability of observing all samples under the parameter θ .

MLE estimates the parameter that maximizes this joint probability.

d) How would using log-likelihood fix computational issues?

Answer :

Using the log-likelihood converts the product of many small probabilities into a sum of logarithms:

$$\log \left(\prod_{i=1}^n P(x_i | \theta) \right) = \sum_{i=1}^n \log P(x_i | \theta)$$

This provides several advantages:

- Prevents numerical underflow caused by multiplying very small probabilities.
- Simplifies mathematical calculations and differentiation.
- Makes optimization algorithms faster and more stable.
- Produces the same optimal parameter estimates because the logarithm is a monotonic increasing function.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

a) List possible implementation errors in the learning algorithm.

Answer :

b) Analyze how incorrect learning rate or weight updates affect convergence.

Answer :

c) Identify dataset-related issues that may cause this problem.

Answer :

d) Propose modifications to ensure convergence.

Answer :

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

a) Identify possible mathematical or coding errors in gradient computation.

Answer :

Possible implementation errors include:

- Incorrect weight update formula.
- Bias term not updated or omitted.
- Wrong prediction or activation function.
- Incorrect class labels (using 0/1 instead of $-1/+1$ without proper handling).
- Errors in feature normalization or data preprocessing.
- Incorrect stopping condition or too few training iterations

b) Explain the role of activation functions in gradient flow.

Answer :

- A very high learning rate causes large weight changes, making the algorithm unstable and preventing convergence.
- A very low learning rate results in slow learning and requires many iterations.
- Incorrect weight updates move the decision boundary in the wrong direction, preventing the Perceptron from finding the correct separating hyperplane even for linearly separable data.

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c) Analyze how vanishing or exploding gradients affect training.

Answer :

Dataset-related issues include:

- Incorrect or noisy class labels.
- Missing or inconsistent feature values.
- Features with very different scales.
- Duplicate or contradictory samples.
- Data that is not truly linearly separable due to preprocessing errors.

d) Suggest techniques to fix the issue (e.g., normalization, initialization).

Answer :

To ensure convergence:

- Use the correct Perceptron update rule:

$$w = w + \eta(y - \hat{y})x$$
- Include and update the bias term.
- Choose an appropriate learning rate (e.g., 0.01–0.1).
- Normalize or standardize input features.
- Verify correct class labels and remove noisy or inconsistent samples.
- Increase the maximum number of training iterations and shuffle the training data each epoch.

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

a) Explain why batch gradient descent is slow in large datasets.

Answer :

Batch Gradient Descent computes the gradient using all training samples before updating the model parameters. For large datasets, each update requires processing the entire dataset, which:

- Increases computation time.
- Requires high memory usage.
- Makes each iteration slow.
- Delays convergence because updates occur only after one complete pass through the data.

b) Analyze the cause of fluctuations in SGD.

Answer : Stochastic Gradient Descent (SGD) updates the model using one training sample at a time. Since each sample provides a noisy estimate of the true gradient:

- The loss fluctuates during training.
- Parameter updates are less stable.
- The optimization path oscillates around the minimum instead of moving smoothly.
- However, these fluctuations can help the model escape local minima and saddle points.

c) Compare the error behavior of Mini-Batch Gradient Descent.

Answer :

Mini-Batch Gradient Descent uses a small batch of samples (e.g., 32, 64, or 128) for each update.

- Error is more stable than SGD because gradients are averaged over several samples.
- It is faster than Batch Gradient Descent since it avoids processing the entire dataset for every update.
- It produces smoother convergence with moderate fluctuations.
- It offers a good balance between computational efficiency and convergence stability.

d) Recommend the best approach and justify with error trade-offs.

Answer :

Mini-Batch Gradient Descent is the best approach because it balances speed, stability, and accuracy.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

a) Explain how high dimensionality increases model error.

Answer :

As the number of features increases, the data becomes sparse, making it difficult for the model to learn meaningful patterns. The model starts fitting noise instead of the underlying relationships, resulting in:

- Increased variance.
- Higher risk of overfitting.
- Poor generalization to unseen data.
- Reduced prediction accuracy on test data.

b) Identify signs of overfitting in this scenario.

Answer :

Signs of overfitting include:

- Very high training accuracy but low testing accuracy.
- Large gap between training and test errors.
- The model memorizes training data instead of learning general patterns.
- Poor performance on new or unseen data.

c) Analyze why distance-based learning fails in high dimensions.

Answer :

Distance-based algorithms (such as KNN and K-Means) rely on measuring distances between data points. In high-dimensional spaces:

- Distances between points become very similar (distance concentration).
- The concept of "nearest neighbor" becomes less meaningful.
- Data points become sparse, making similarity measures unreliable.
- As a result, classification and clustering accuracy decrease.

d) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer :

To improve performance:

- Apply Principal Component Analysis (PCA) to reduce the number of features.
- Use Linear Discriminant Analysis (LDA) for supervised dimensionality reduction.
- Perform feature selection to remove irrelevant or redundant features.
- Apply L1 (Lasso) or L2 (Ridge) regularization to reduce overfitting.
- Increase the amount of training data and use cross-validation for better model generalization.
- Use dropout (for neural networks) to prevent overfitting.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand